

# Scalable Transpilation for Overcoming Restricted Connectivity in Distributed Superconducting Quantum Architectures

Afonso Azenha<sup>1,2</sup>, Ilia Polian<sup>1,2</sup>, Sebastian Brandhofer<sup>3</sup>

<sup>1</sup>Institute of Computer Architecture and Computer Engineering, University of Stuttgart

<sup>2</sup>Center for Integrated Quantum Science and Technology (IQST)

<sup>3</sup>IBM Quantum, IBM Research Europe

## Core Contributions

- Contribution 1: SABRE Adaptations for Distributed Quantum Computing (DQC):** Introduces a fast routing heuristic tailored for the restricted connectivity of distributed architectures.
- Contribution 2: Scalable Partitioning-based Mapping:** Replaces complex state-of-the-art hypergraphs with **lightweight Quantum Interaction Graphs (QIGs)** for efficient qubit-to-QPU assignment.
- Overhead Reduction:** Pairing the QIG mapping with our DQC-adapted SABRE variants yields an **end-to-end transpilation framework** that cuts routing costs by up to 52% against a DQC-adapted SABRE baseline, and up to 24% over *DMapS* (a state-of-the-art DQC transpiler).
- Demonstrated Scalability:** Validated on distributed architectures up to **399 qubits**, delivering **> 250 ×** speed-ups over *DMapS*.

## The DQC Transpilation Bottleneck

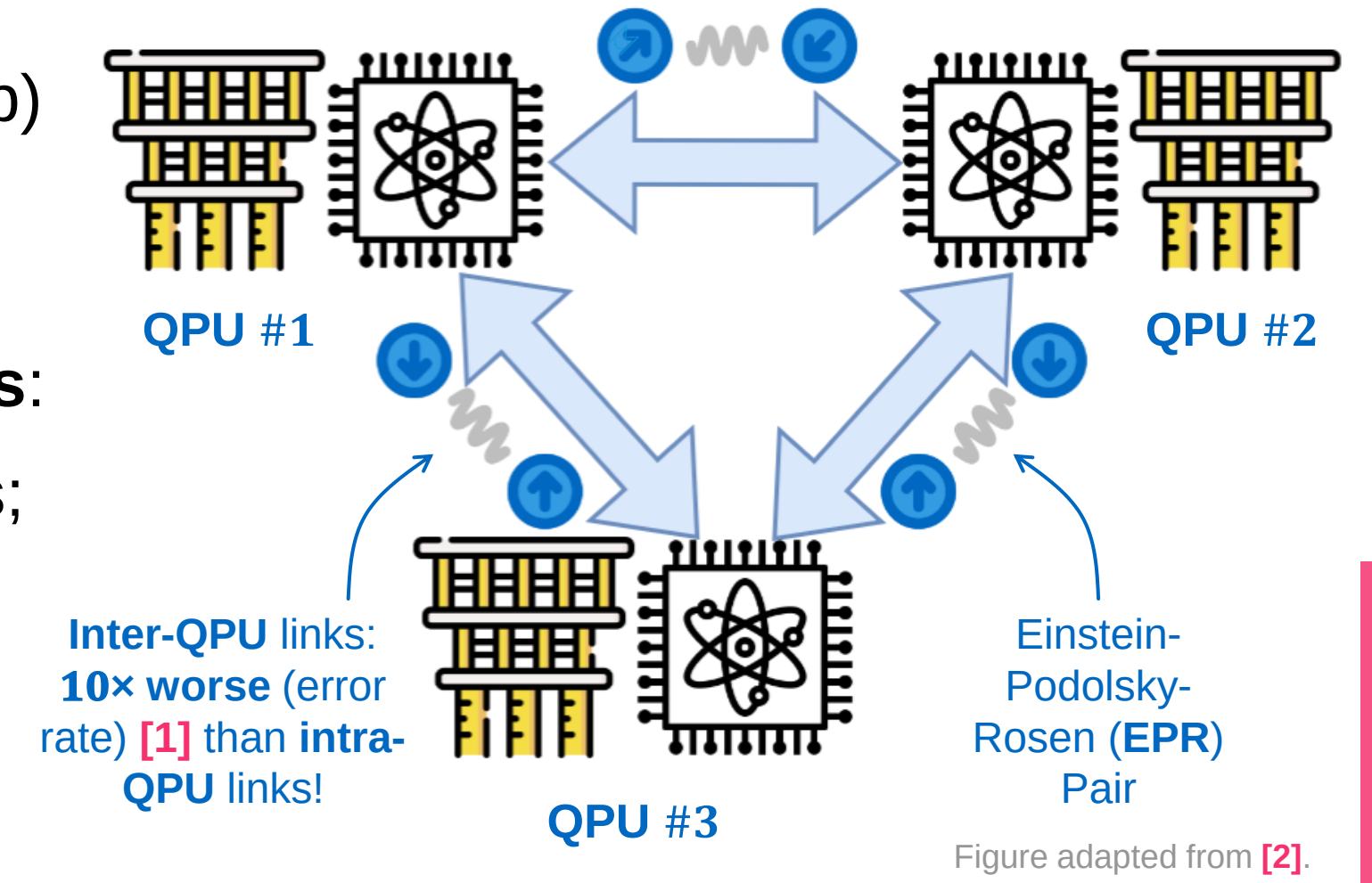
- High Communication Cost:** Inter-module links exhibit high error rates, making remote operations a primary optimization target.
- Monolithic Transpilers** (e.g., *SABRE* [4], *tket* [5]): don't take into account DQC-specific information ⇒ **unsuited for DQC**;
- DQC-specific Transpilers** (e.g., *pytket-dqc* [3], *DMapS* [6]):
  - Assume idealized, **all-to-all intra-module connectivity**;
  - Suffer from **severe runtime overheads** on larger systems.

⇒ **Our Goal:** Highly performant and scalable transpilers for (superconducting) DQC!

## Distributed Superconducting Quantum Computing

### Motivation

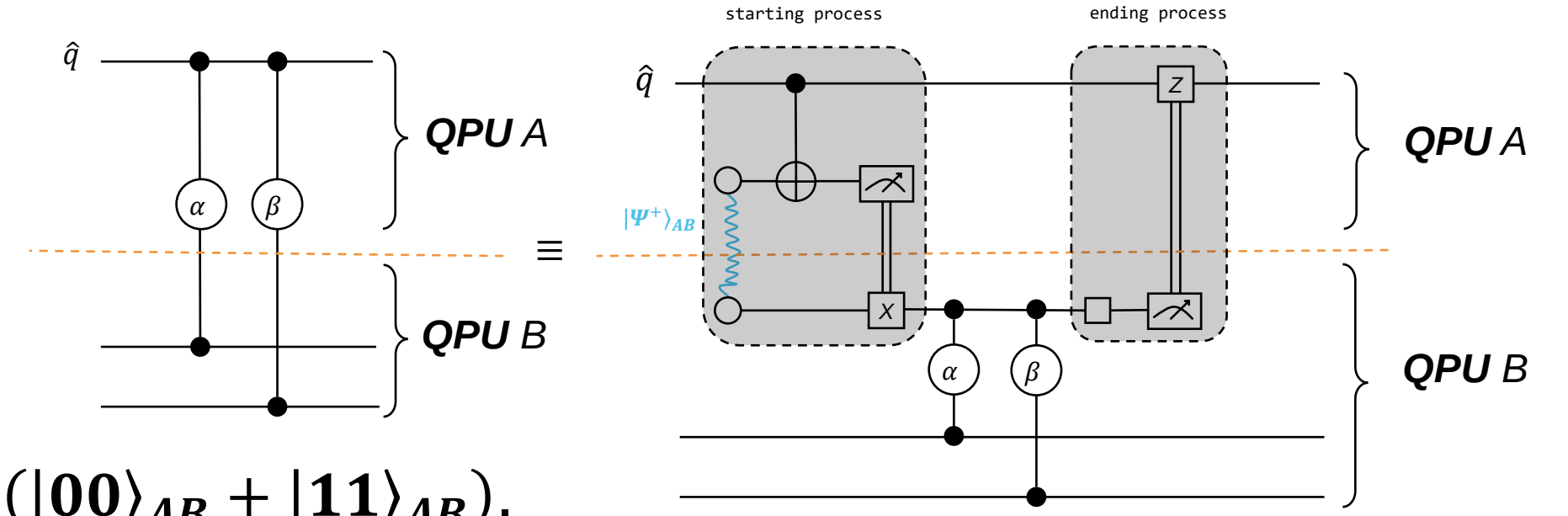
- Scaling **monolithic** (single-chip) superconducting quantum processing units (QPUs) faces **major engineering challenges**:
  - Complex control electronics;
  - Cryogenic cooling power requirements;
  - Decreasing manufacturing yields.



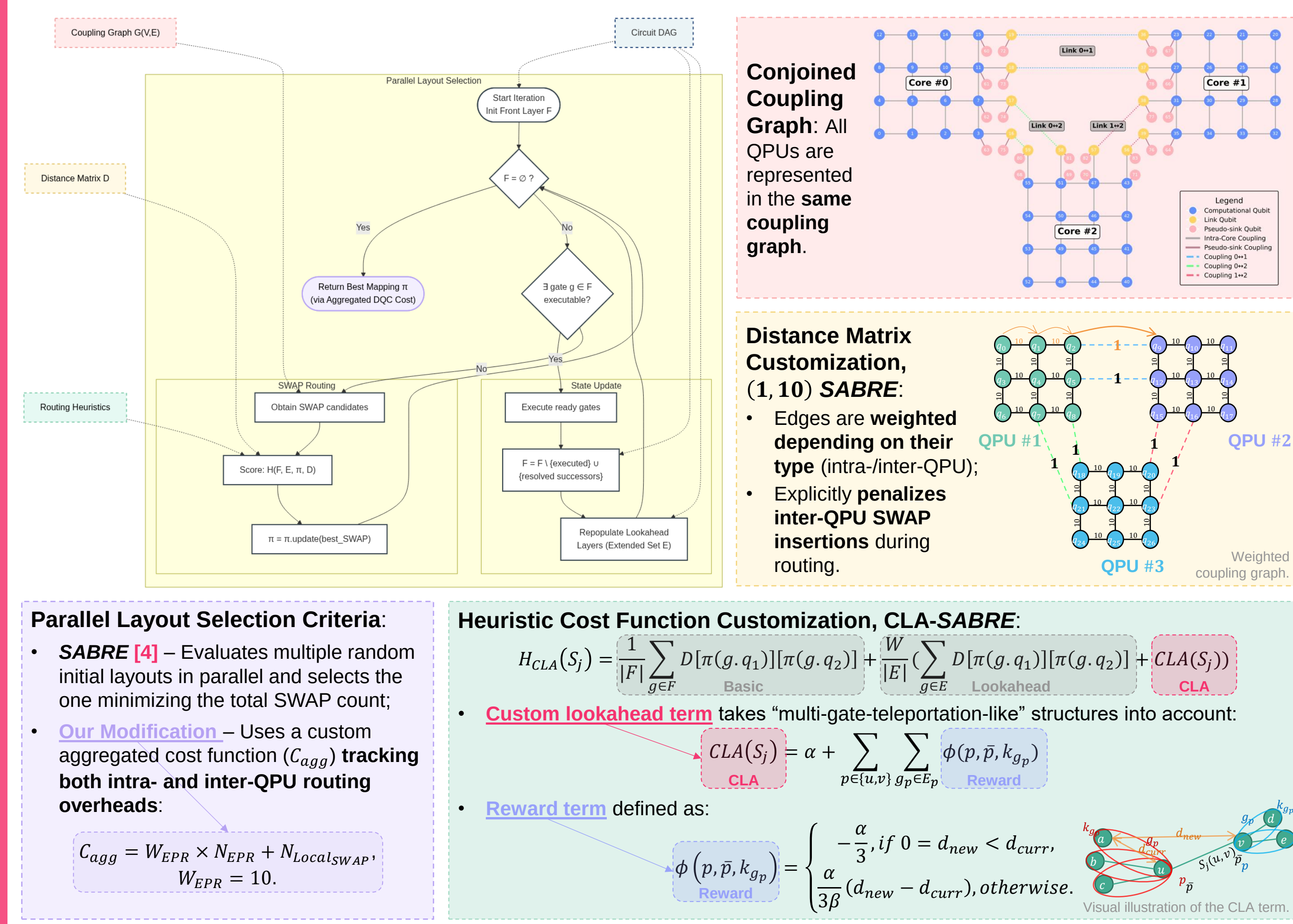
- Alternative:** Multi-QPU architectures for distributed quantum computing (DQC) ⇒ Allows to **scale the number of qubits!**

### Background & Assumptions

- Remote quantum links implemented via **quantum gate teleportation** mediated by **EPR pairs**:  $|\Psi^+\rangle_{AB} = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$ .
- Max. 1 shared EPR pair per inter-QPU link.
- Quantum gate teleportation only!** ⇒ 1 inter-QPU SWAP  $\equiv$  3 EPR pairs.



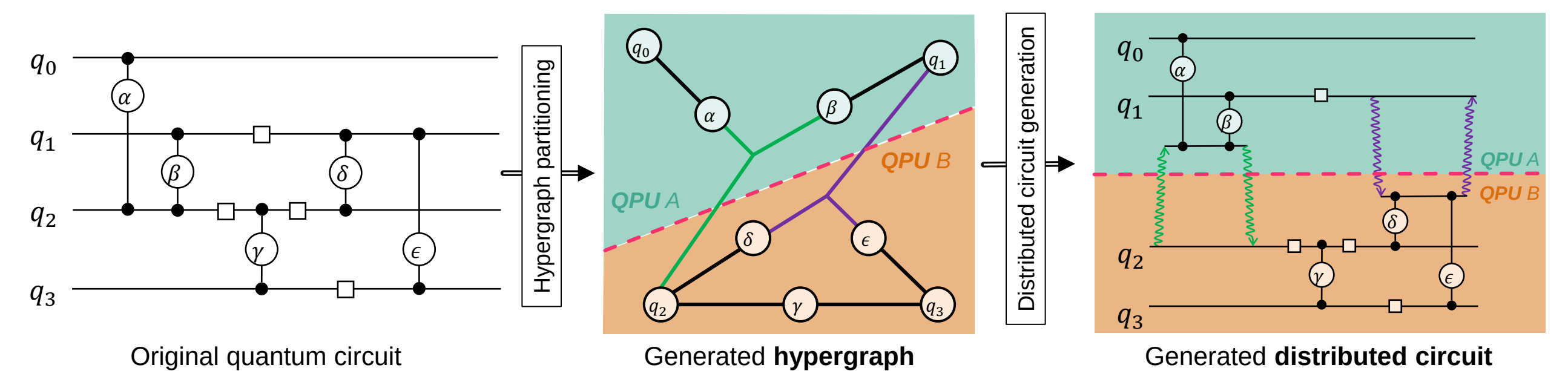
## Contribution 1: SABRE Adaptations for DQC



## Contribution 2: Partitioning-based Mapping

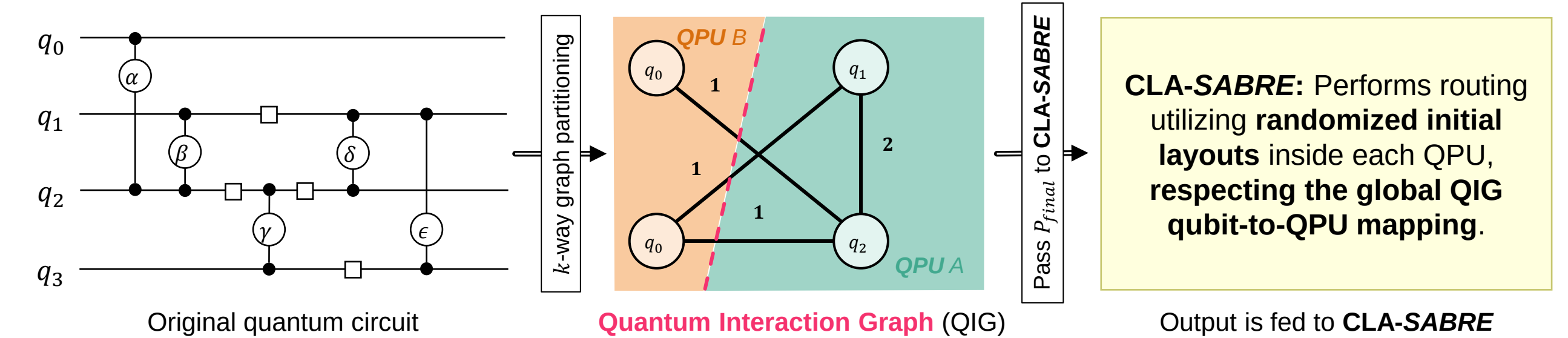
### State-of-the-Art: Hypergraph Partitioning

- State-of-the-Art Baseline** (*pytket-dqc* [3]): Utilizes hypergraphs to represent quantum circuits, where hyperedges model “distributable packets” in multi-gate teleportation.
- Qubit-to-QPU Assignment:** Minimizes remote communication overhead by computing high-quality qubit-to-QPU assignments via hypergraph partitioning.
- Scalability Bottleneck:** Suffers from high computational complexity, making it prohibitive for large-scale circuits.



### Our Proposal: Reducing Complexity by Replacing Hypergraphs with Quantum Interaction Graphs

- Trade-off for Scalability:** Reduces representation size and complexity compared to hypergraphs by omitting explicit multi-gate teleportation (hyperedge) tracking.
- Initial Qubit Assignment:** Efficiently determines the global mapping of virtual qubits to target physical QPUs without prohibitive runtime overhead.
- End-to-End Integration:** Establishes a topology-aware foundation for DQC-modified SABRE routines to efficiently route local and inter-core gates.



## Results (Benchmarking)

→ **Key Results:** QIG + CLA-SABRE eliminates hypergraph timeouts (T.O.) at utility scale (399 qubits), yielding up to 52% cost reduction and > 250 × speed-ups. (Superscripts *a–d* denote partial timeouts where some circuits failed to compile within the 1-hour or 3-hour limits for the 48-qubit and 399-qubit architectures).

Geometric Mean of Aggregated Cost Reduction and Runtime Speed-up Across DQC Architectures

| Algorithm                       | 48-Qubit (All-to-All) |                      |                     |                      | 399-Qubit IBM Flamingo (All-to-All) |            |                     |                     | 399-Qubit IBM Flamingo (Line) |            |                     |                     |
|---------------------------------|-----------------------|----------------------|---------------------|----------------------|-------------------------------------|------------|---------------------|---------------------|-------------------------------|------------|---------------------|---------------------|
|                                 | Unstructured          | Structured           | Unstructured        | Structured           | Unstructured                        | Structured | Unstructured        | Structured          | Unstructured                  | Structured | Unstructured        | Structured          |
| DMapS                           | 0.00%                 | 1.00×                | 0.00%               | 1.00×                | T.O.                                | T.O.       | 44.90% <sup>d</sup> | 0.002× <sup>d</sup> | T.O.                          | T.O.       | 47.43% <sup>d</sup> | 0.002× <sup>d</sup> |
| (1,10) SABRE                    | −0.54%                | 393.313×             | −68.18%             | 296.675×             | 10.30%                              | 0.908×     | −93.10%             | 0.764×              | 5.92%                         | 0.881×     | −72.64%             | 0.763×              |
| CLA-SABRE ( $ E  = 20$ )        | 8.93%                 | 245.031×             | −21.61%             | 264.429×             | −127.79%                            | 0.095×     | −106.95%            | 0.350×              | −261.50%                      | 0.032×     | −43.47%             | 0.184×              |
| CLA-SABRE ( $ E  = 100$ )       | 7.54%                 | 125.559×             | −36.53%             | 74.640×              | −52.62%                             | 0.053×     | −58.43%             | 0.162×              | −143.70%                      | 0.032×     | −58.43%             | 0.143×              |
| pytket-dqc (CE) + SABRE         | 33.05% <sup>a</sup>   | 0.428 × <sup>a</sup> | T.O. <sup>b</sup>   | T.O. <sup>b</sup>    | T.O.                                | T.O.       | T.O.                | T.O.                | T.O.                          | T.O.       | T.O.                | T.O.                |
| pytket-dqc (PH) + SABRE         | 23.53% <sup>a</sup>   | 0.256 × <sup>a</sup> | T.O. <sup>b</sup>   | T.O. <sup>b</sup>    | T.O.                                | T.O.       | T.O.                | T.O.                | T.O.                          | T.O.       | T.O.                | T.O.                |
| HGP + CLA-SABRE ( $ E  = 20$ )  | 26.70%                | 24.348×              | 28.94% <sup>c</sup> | 19.152× <sup>c</sup> | T.O.                                | T.O.       | T.O.                | T.O.                | T.O.                          | T.O.       | T.O.                | T.O.                |
| QIG + SABRE                     | −8.90%                | 281.934×             | 16.87%              | 403.934×             | 0.35%                               | 1.029×     | 46.99%              | 1.051×              | 1.68%                         | 1.182×     | 48.41%              | 1.264×              |
| QIG + (1,10) SABRE              | 9.75%                 | 192.773×             | 5.34%               | 291.622×             | 10.07%                              | 0.899×     | 40.33%              | 0.977×              | 7.21%                         | 1.073×     | 43.35%              | 1.269×              |
| QIG + CLA-SABRE ( $ E  = 20$ )  | 24.72%                | 139.597×             | 24.01%              | 252.574×             | −126.62%                            | 0.090×     | 47.64%              | 0.734×              | −243.71%                      | 0.038×     | 48.80%              | 0.858×              |
| QIG + CLA-SABRE ( $ E  = 100$ ) | 15.95%                | 70.055×              | 5.44%               | 71.020×              | −52.62%                             | 0.053×     | 51.94%              | 0.331×              | −134.55%                      | 0.035×     | 51.96%              | 0.373×              |

### References

- [1] K. Heya et al., “Randomized benchmarking of a remote CNOT gate via a meter-scale microwave link”, 2025.
- [2] R. Zhou et al., “Optimizing Compilation for Distributed Quantum Computing via Clustering and Annealing”, 2025.
- [3] P. Andres-Martinez et al., “Distributing circuits over heterogeneous, modular quantum computing network architectures”, 2024.
- [4] G. Li et al., “Tackling the qubit mapping problem for NISQ-era quantum devices,” 2019.
- [5] S. Sivarajah et al., “tket: a retargetable compiler for NISQ devices”, 2020.
- [6] T. Luo et al., “DMapS: End-to-end qubit mapping and routing for distributed quantum computing architectures”, 2025.
- [7] J. Eisert et al., “Optimal local implementation of nonlocal quantum gates”, 2000.